Script-Controlled ACL – Restrict Record Access Based on Field Value

PERFORMANCE TESTING

Script-Controlled ACL Record-Level Security
ServiceNow Access Control (ACL)

Date:	October 03, 2026	**Maximum Marks:**	4 Marks
Team ID:	SWTID-2026-8089	**Project Name:**	Script-Controlled ACL

1. Test Overview

Project Name:	Script-Controlled ACL – Restrict Record Access Based on Field Value
Test Type:	Functional performance / ACL execution observation
Application:	ServiceNow
Test Date:	October 03, 2026

This performance-testing document evaluates the observed execution of the Script-Controlled ACL solution. The objective is to verify that the configured ServiceNow ACLs correctly allow or deny read, create, write and delete access to the Institution Details records based on the user's roles and the Branch field value.

2. Testing Scope

- Role assignment to the EEE User (bb1, bb2, bb3, bb4)
- READ ACL evaluation (role, data condition Branch is EEE, advanced script)
- Record filtering for the EEE User, a user without roles and the administrator
- CREATE, WRITE and DELETE ACL enforcement
- End-to-end access-control verification through impersonation

3. Test Environment

Platform: ServiceNow instance used for the project

Feature: Access Control (ACL) + Users & Roles

Table: Institution Details (u_institution_details)

Roles: bb1 (read), bb2 (create), bb3 (write), bb4 (delete)

No credentials are recorded in this document.

4. Performance / Execution Test Cases

Test ID	Scenario	Execution	Expected Observation	Observed Result	Status
PT-001	EEE User read access	Impersonate the EEE User (bb1) and open the Institution Details list.	Only EEE branch records should be displayed.	Only the EEE branch record was displayed.	**PASS**
PT-002	User without roles	Impersonate a user without bb1–bb4 and open the list.	No records should be visible.	Access was denied; no records were displayed.	**PASS**

Test ID	Scenario	Execution	Expected Observation	Observed Result	Status
PT-003	Administrator access	Open the list as the administrator.	All records should be visible regardless of branch.	Records of all branches (ECE, EEE, CSE) were displayed.	**PASS**
PT-004	Create access (bb2)	Check the list as a user holding bb2.	The New button should be available.	The New button was available for the permitted user.	**PASS**
PT-005	Write access (bb3)	Edit the Description of an EEE record.	The record should be updated.	The Description was updated successfully.	**PASS**
PT-006	Delete access (bb4)	Select an EEE record, choose Delete and confirm.	The record should be deleted.	The record was deleted after confirmation.	**PASS**
PT-007	Measured response-time benchmark	Measure the elapsed time to load the list with ACL evaluation using a consistent timing method.	A measured response time should be recorded for benchmarking.	No controlled timing measurement was captured in the existing implementation evidence.	**NOT MEASURED**

5. Performance Observations

- The configured ACLs completed the intended access-control path in the recorded implementation.
- The READ ACL restricted bb1 users to EEE branch records, while users without the roles saw no records and the administrator saw all records.
- The CREATE, WRITE and DELETE ACLs enforced access through the bb2, bb3 and bb4 roles.
- The available evidence demonstrates functional ACL execution, but it does not provide controlled response-time, throughput, concurrency, or load-test measurements.

6. Performance Metrics

Metric	Result	Evidence / Note
ACL evaluation (role + Branch condition)	Successful	EEE User (bb1) saw only EEE branch records.
Access denial for users without roles	Successful	No records were displayed to the user without roles.
Administrator access	Successful	All records were displayed to the administrator.
Create / Write / Delete enforcement	Successful	Operations were verified for the bb2, bb3 and bb4 roles.
Response time	Not measured	No controlled timing data was captured.
Throughput	Not measured	No load/concurrency test was performed.
Concurrent users	Not measured	No concurrency test was performed.

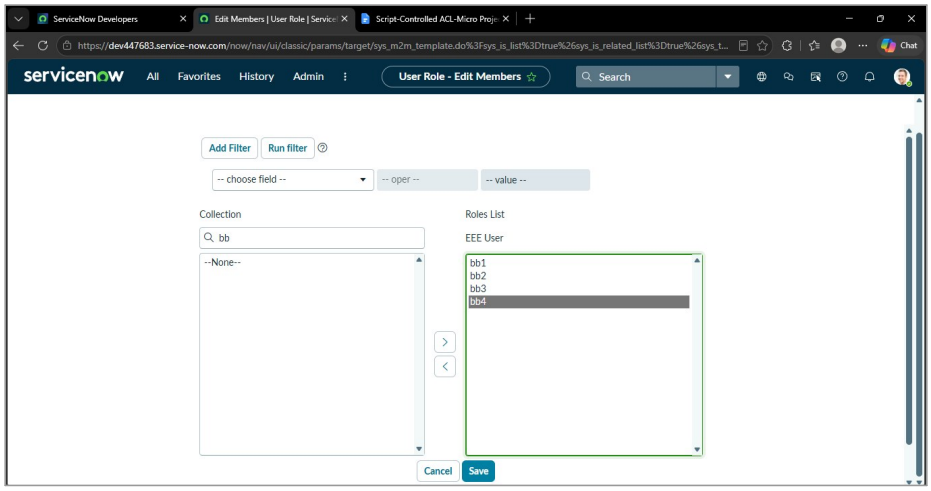
7. Test Result

The recorded implementation successfully demonstrated the intended functional execution of the Script-Controlled ACL solution. The ACLs allowed access only to users who met the role and Branch condition, denied access to users without the roles, and preserved full access for the administrator.

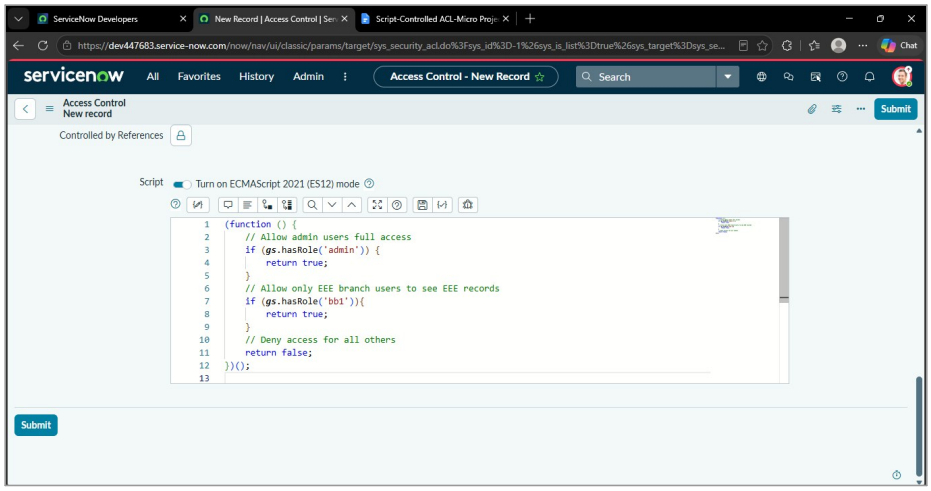
Quantitative performance benchmarks such as response time, throughput, and concurrent-user capacity were not measured in the available project evidence and therefore are not reported as measured results.

8. Evidence to Attach

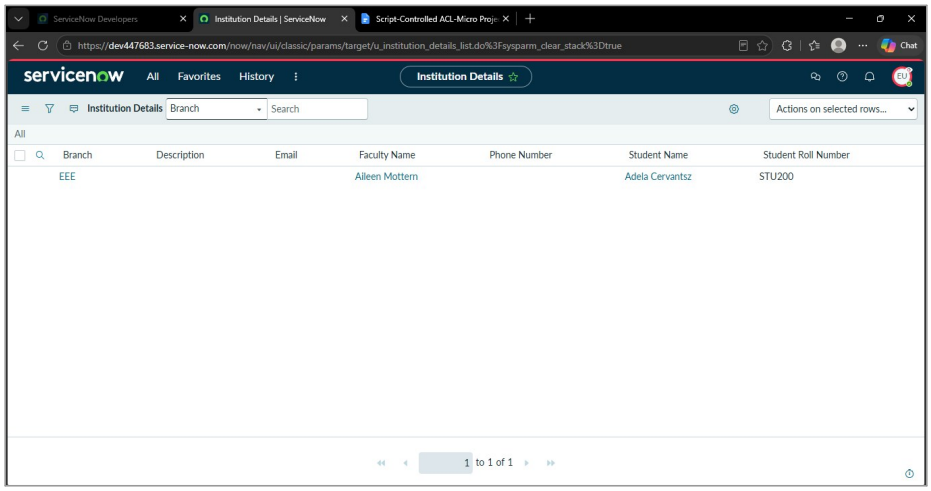
- Roles bb1–bb4 assigned to the EEE User
- Read ACL configuration (role bb1, Branch is EEE, advanced script)
- EEE User view – only EEE branch records
- User without roles – access denied
- Administrator view – all records
- Create, Write and Delete ACL configuration and verification
- Demo video link (to be added by the team)



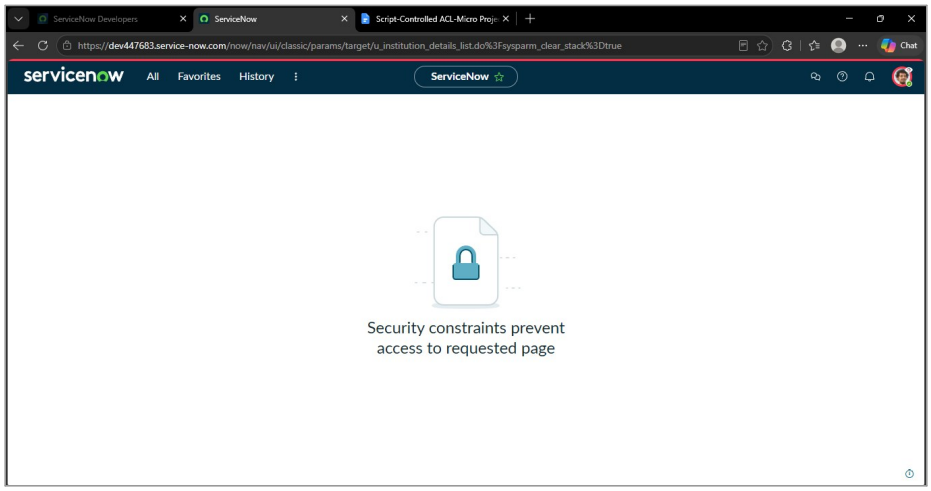
Evidence 1: Roles bb1–bb4 assigned to the EEE User



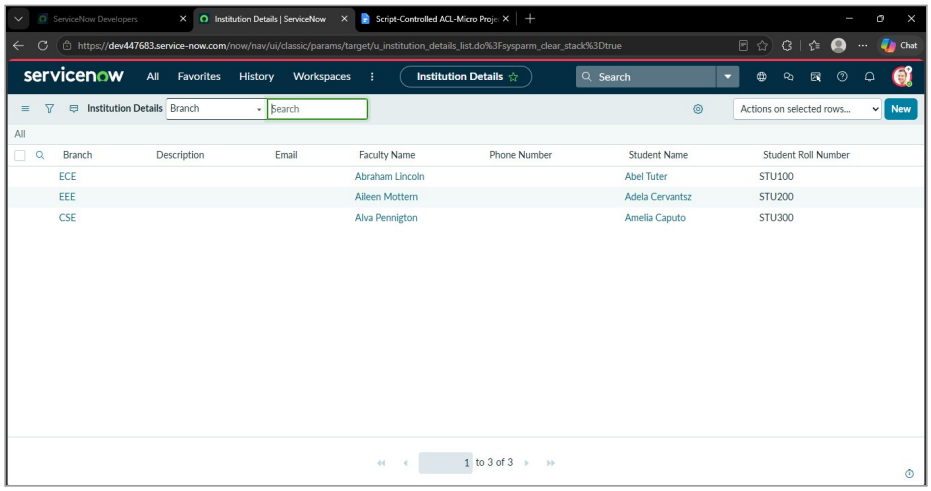
Evidence 2: Read ACL – advanced script



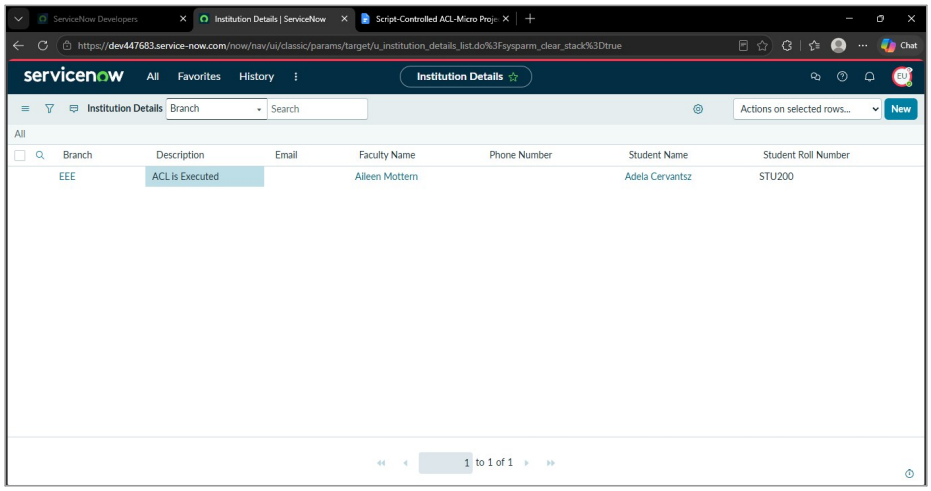
Evidence 3: EEE User (bb1) views only EEE branch records



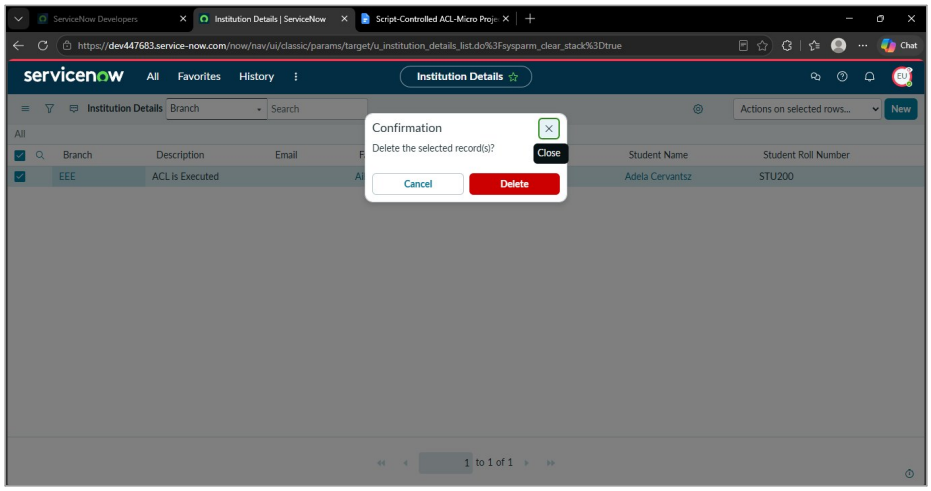
Evidence 4: User without roles – security constraints prevent access



Evidence 5: Administrator views all records



Evidence 6: Record edited with write access



Evidence 7: Delete confirmation for the selected EEE record

9. Conclusion

Based on the available execution evidence, the Script-Controlled ACL solution completed its intended access-control workflow and produced the expected allow and deny outcomes for each user type. Additional quantitative load or response-time benchmarking would require a controlled test run with timing and concurrency measurements.